

Literature resolution of the general-tree 2-Steiner distance-matrix rank problem

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Status. The open problem at the end of arXiv:2108.12798 was answered in full in 2022 by Ali Azimi and Sivaramakrishnan Sivasubramanian. This packet records the exact correspondence and claims no new mathematical result.

Definition. Let T be a tree with vertex set $V(T)$. If $S \subseteq V(T)$, its Steiner distance $d_T(S)$ is the number of edges in the smallest connected subtree containing S . The 2-Steiner distance matrix $D_2(T)$ is indexed by the 2-element subsets of $V(T)$, with

$$D_2(T)_{P,Q} = d_T(P \cup Q).$$

Exact source question. Azimi, Bapat, and Goel prove in arXiv:2108.12798 that if C is a caterpillar on N vertices with p pendant vertices, then

$$\text{rank } D_2(C) = 2N - p - 1.$$

They end the paper with the following question.

$$= 2n - 1 + \sum_{j=1}^{n-1} i_j,$$

the proof is complete. □

We conclude this paper with the following open problem.

Open Problem: If T is a general tree on n vertices, find $\text{rank}(D_2(T))$.

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Exact later answer. The first theorem of Azimi and Sivasubramanian's 2022 paper treats an arbitrary tree, not merely a caterpillar.

Published answer. *Let T be a tree on n vertices having p pendant vertices (leaves). Then*

$$\text{rank } D_2(T) = 2n - p - 1.$$

of the two matrices M_T and N_T . That is, it is the average of the min 4PC and the max 4PC matrices. Thus, we also use the term *average 4PC matrix* for the matrix $\mathfrak{D}_2(T)$. It is interesting to note that among the min 4PC, the max 4PC and the average 4PC matrix, we only know results about the min 4PC matrix. This gives another motivation to study the average 4PC matrix or $\mathfrak{D}_2(T)$.

As we will see, several similarities and differences exist among our results on the 2-Steiner distance matrix $\mathfrak{D}_2(T)$ and results about the distance matrix D_T of T . Our first result is the following.

Theorem 1. *Let T be a tree on n vertices having p pendant vertices (or leaves). Then,*

$$\text{rank}(\mathfrak{D}_2(T)) = 2n - p - 1.$$

We give a proof of Theorem 1 in Section 3. While determining the rank of $\mathfrak{D}_2(T)$, we explicitly give a basis $\mathfrak{B}$ for the row space of $\mathfrak{D}_2(T)$. Graham and Lovasz in [17] gave an explicit formula for the inverse of D_T . They showed that the inverse of D_T

Thus the caterpillar formula survives unchanged for every tree and answers the source’s literal request.

Proof route in the answer paper. The published proof is by induction on the number of vertices. After the three-vertex base case, it separates two reductions. If a vertex supports at least two leaves, deleting one leaf removes one row-space direction; if every leaf has a degree-two neighbour, deleting a terminal two-vertex path removes two directions. The leaf count changes in exactly the corresponding way, so the induction preserves the expression $2n - p - 1$. The proof also constructs an explicit basis for the row space. Later sections compute an inverse for the nonsingular restriction to that basis, its determinant, and the inertia of $D_2(T)$; these strengthen, but are not needed for, the rank answer.

Verification. The source PDF included here contains the displayed open problem on its page 23. The answer PDF is the published version in *Linear Algebra and its Applications*; its abstract states the formula and Theorem 1 on journal page 67 gives it for every tree. Its proof appears in Section 3. The objects and parameters agree exactly: n is the order of the tree, p is its number of pendant vertices, and the matrix is indexed by vertex pairs with entries equal to the Steiner distance of their union.

Conclusion and limitation. The target is classified as `literature_already_answered`. No novelty is claimed for the formula or proof. Human review should verify only the bibliographic attribution and the exact notational match recorded above.

References.

1. A. Azimi, R. B. Bapat, and S. Goel, *Steiner distance matrix of caterpillar graphs*, Special Matrices 10 (2022), 167–188; arXiv:2108.12798, doi:10.1515/spma-2022-0162.
2. A. Azimi and S. Sivasubramanian, *The 2-Steiner distance matrix of a tree*, Linear Algebra Appl. 655 (2022), 65–86, doi:10.1016/j.laa.2022.09.007.